

GENERATIVE AI PROJECT 3

Multimodal Image Generation Studio

Submitted By

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1. Project Title

Multimodal Image Generation Studio

2. Objective

The objective of this project is to develop a Python-based application that converts natural language text descriptions into AI-generated images. The application interacts with an image generation API, accepts user prompts, allows image size selection, generates images, and saves them locally for future use.

3. Problem Statement

Creating digital artwork manually requires significant time and artistic skills. Generative AI enables users to create high-quality images simply by describing them in natural language. This project demonstrates how AI-powered image generation can be integrated into Python applications.

4. Technologies Used

- Python 3
- Requests Library
- Pollinations AI Image Generation API
- Visual Studio Code

5. Features

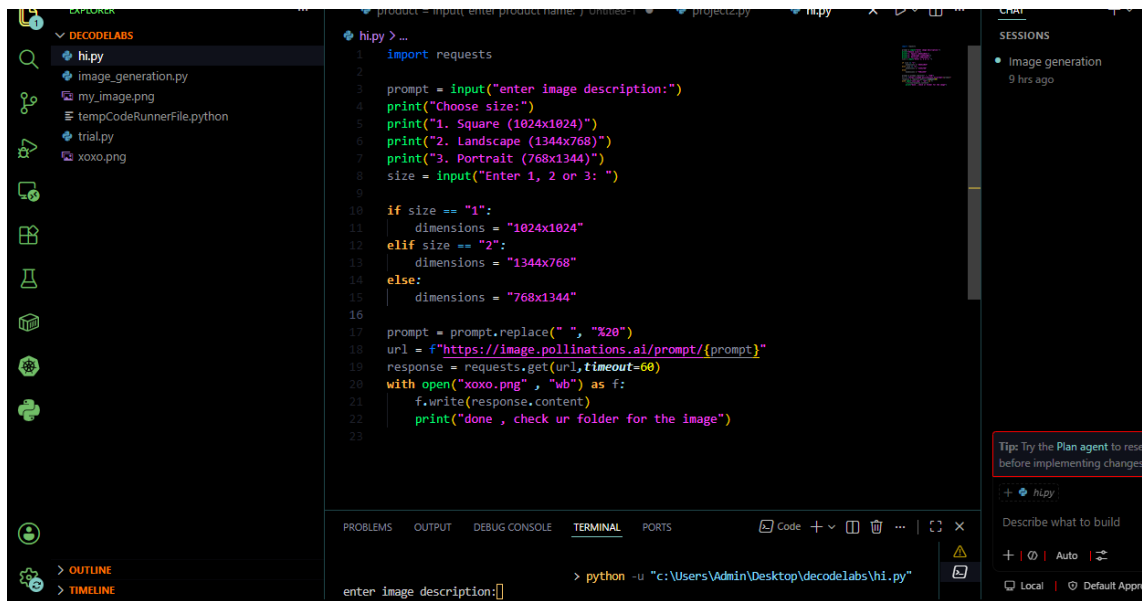
- Accepts image descriptions from users.
- Allows selection of image dimensions.
- Generates AI-based artwork from text prompts.
- Downloads and saves generated images automatically.
- Simple command-line interface.

6. Project Workflow

1. User enters an image description.
2. User selects image size.
3. Application sends request to image generation API.
4. API generates image.
5. Generated image is downloaded.
6. Image is saved locally on the system.

7. Source Code

The project is implemented in Python using the Requests library to communicate with the image generation API and save the generated image.



8. Test Cases

Test Case 1

Input Prompt:

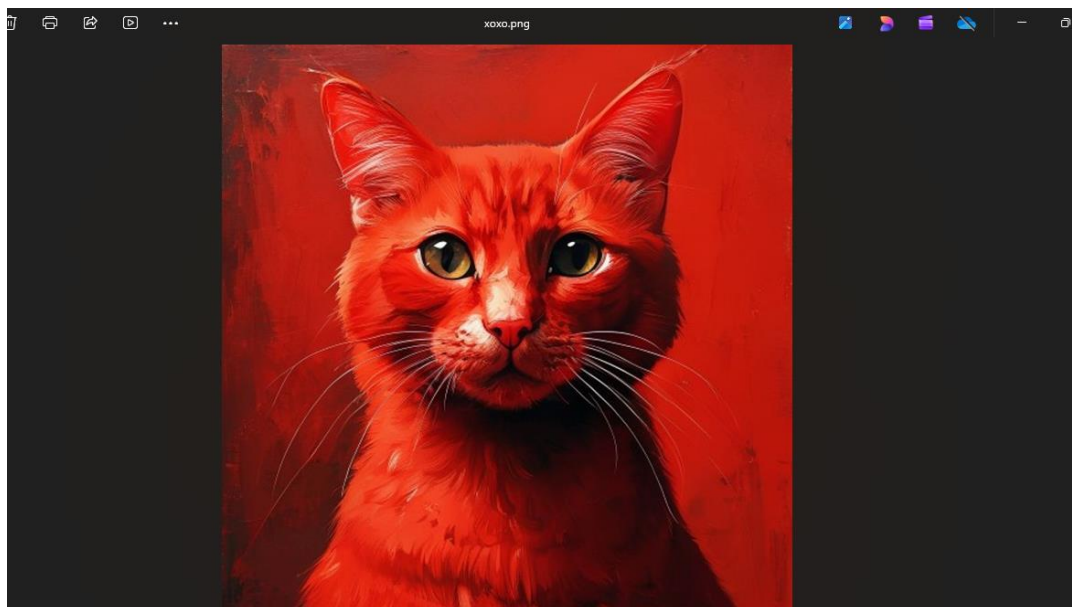
Red Cat

Expected Result:

AI-generated image of a red-colored cat.

Status:

Passed



Test Case 2

Input Prompt:

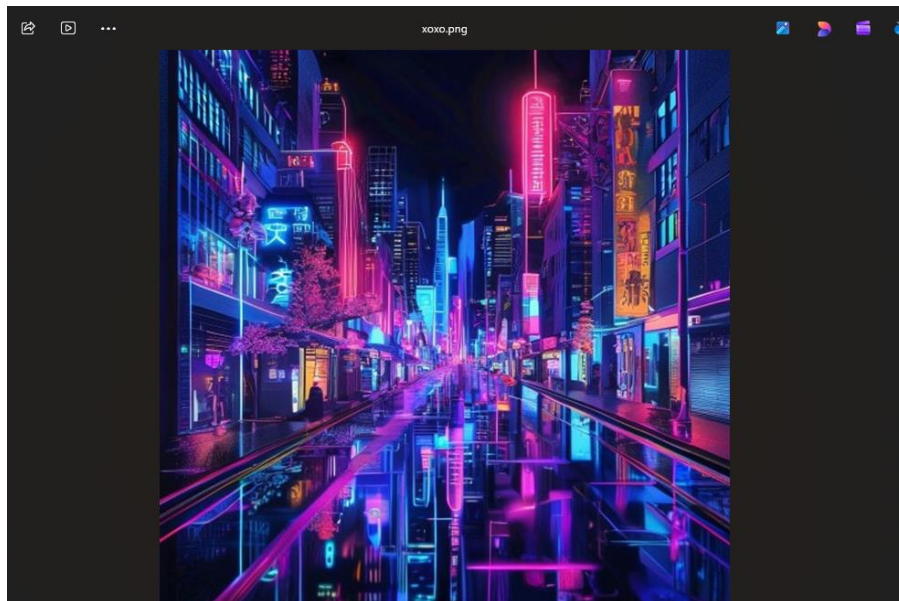
Neon City

Expected Result:

AI-generated futuristic city with neon lights.

Status:

Passed



Test Case 3

Input Prompt:

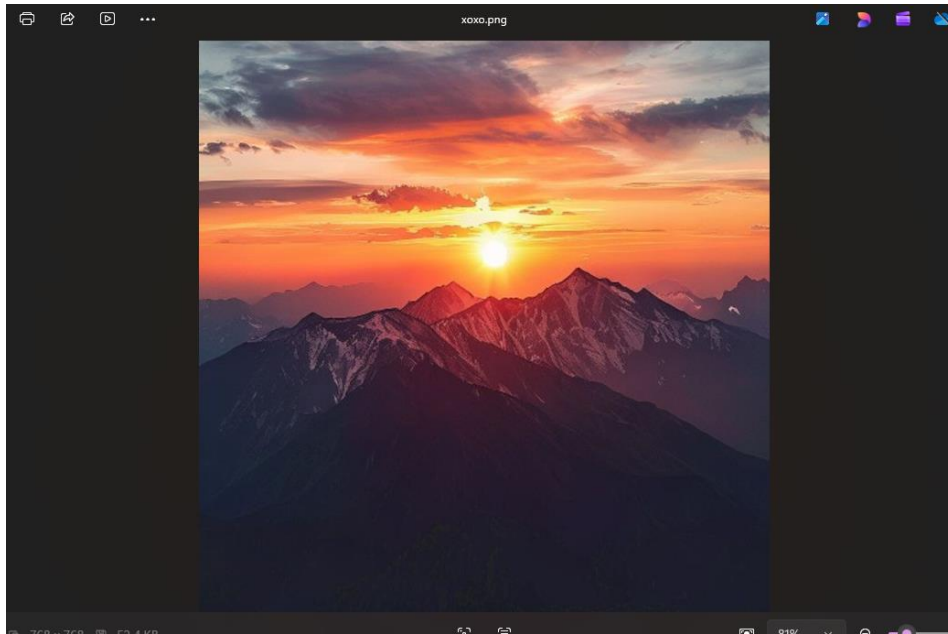
Sunset Over Mountains

Expected Result:

AI-generated mountain landscape during sunset.

Status:

Passed



9. Conclusion

This project successfully demonstrates the integration of a Generative AI Image API with Python. The application converts user text prompts into visual artwork and saves the generated images locally. Through this project, practical experience was gained in API integration, image handling, and Generative AI application development.

10. References

1. Python Documentation
2. Requests Library Documentation
3. Pollinations AI API Documentation
4. DecodeLabs Industrial Training Kit